M1 — Anchor Paper Notes (EMG baseline)

Anchor paper

Rahimi, Benatti, Kanerva, Benini, Rabaey — "Hyperdimensional Biosignal Processing: A Case Study for EMG-based Hand Gesture Recognition." IEEE International Conference on Rebooting Computing (ICRC), San Diego, Oct 2016, pp. 1–8.

- **PDF (authoritative, ETH host):** https://iis-people.ee.ethz.ch/~arahimi/papers/ICRC16.pdf
- **Publication record / DOI:** https://researchr.org/publication/RahimiBKBR16 (IEEE Xplore DOI: 10.1109/ICRC.2016.7738683)
- **Authors' released code + data:** https://github.com/abbas-rahimi/HDC-EMG

Chosen because it is the most-cited HDC + EMG work, reviewers know its numbers, and the authors released code + data (so an exact reproduction is possible).

Headline numbers to reproduce

Configuration	Reported accuracy
Spatial encoding only (N = 1, sample-by-sample)	**90.8%** (avg over 5 subjects)
Spatiotemporal encoding (best N-gram per subject)	**97.8%** (avg over 5 subjects)
SVM baseline (N = 1)	89.7%

Dimensionality robustness: full accuracy held down to **D = 6000**; ~96% down to D = 300; collapses to ~62% below that. (Original D = 10 000.)

Our May target: reproduce the **spatial-only ≈ 90.8%** number first (simplest, no N-gram), then the spatiotemporal ≈ 97.8% number.

Exact protocol in the paper

Parameter	Paper value
HDC model	**MAP**: bipolar {−1, +1}, bind = point-wise multiply, bundle = addition, permute = cyclic shift by 1
Dimension D	**10 000**
Similarity	**Cosine**
Channels	4 EMG sensors (forearm flexor/extensor muscles)
Classes	**5**: closed hand, open hand, 2-finger pinch, point index, rest

Parameter	Paper value
Sampling	**500 Hz**
Input to encoder	**Enveloped EMG, discretized to 21 levels** (0–20 mV, 1 mV step). NOT Hudgins features.
Item memory (iM)	4 orthogonal channel hypervectors
Continuous iM (CiM)	21 level hypervectors; flip D/2/20 bits per step (continuous mapping)
Spatial record	$R[t] = \Sigma_c \, iM(CH_c) * CiM(S_c[t])$
Temporal	N-gram via permutation: $\text{N-gram}[t] = \Pi_{i=0}^{N-1} \rho^i R[t-i]$, $N \in [1,10]$
Best N per subject	S1=4, S2=4, S3=3, S4=5, S5=4 (Table I); downsampling 250 (S5=50)
Training rule	Conditional add: GV(label) += R[t] only if $\cos(GV, R[t]) < 0.9$
Train fraction	**25%** of dataset, uniform random sampling
Subjects	**5** (subset of the 9-subject Benatti 2014 dataset)

Dataset is Benatti et al. 2014 (9 subjects), NOT the UCI Lobov set. 10 repetitions × 3 s contraction + 3 s rest per gesture.

⚠️ Gaps between this paper and our plan / hardware

The earlier research-plan draft assumed a few things that do **not** match Rahimi 2016. Flagging them now so the May reproduction is honest:

Aspect	Rahimi 2016	Our plan / RTL	Action
HDC model	MAP, bipolar	**BSC, binary {0,1}**, bind = XOR	Reproduce in MAP first for parity, then re-derive a BSC binary number as *our* baseline
Dimension	10 000	**1024**	Expected lower accuracy at 1024; D-sweep is Hook A anyway
Similarity	Cosine	**Hamming / popcount**	Cosine for parity; Hamming is the binary analogue
Features	21-level enveloped signal	Plan said **Hudgins (MAV/RMS/WL/ZC/SSC), 16 levels**	Use envelope discretization to match the paper; revisit Hudgins later
Dataset	Benatti 2014 (9 subj)	Plan said **UCI Lobov**	Use Rahimi's released data for exact repro; keep UCI as secondary
Training	Conditional add (cos < 0.9)	Plain majority bundle	Implement conditional add to match

Conclusion: to literally "reproduce a published number," use the authors' released MAP/D=10000/cosine setup on their data. Then convert to *our* BSC / D=1024 / Hamming model and record that as the project's own baseline — the gap between the two is itself a useful, honest result for the paper.

Reproduction strategy (feeds M2–M17)

1. **Stage A — literal parity (MAP):** clone `abbas-rahimi/HDC-EMG`, run their pipeline, confirm ≈90.8% (spatial) / ≈97.8% (spatiotemporal). This is the "published number" checkpoint.
2. **Stage B — our binary model:** re-implement the same encoding in `hdc_ref.py` as BSC (XOR bind, majority bundle, Hamming), D ∈ {1024, ...}, on the same data. Record accuracy as the **project baseline**.
3. **Stage C — config freeze:** lock seed, D, levels, N, train fraction, CV in a config file (M16/M17).

`hdc_ref.py` already provides BSC bind/permute/bundle/classify and continuous item memory, so Stage B reuses existing code. Stage A needs the authors' repo (MATLAB/Python) and their data.

Open questions to resolve in M2–M4

- Is `abbas-rahimi/HDC-EMG` data redistributable, or must the Benatti 2014 set be requested separately? (Check repo license before committing data.)
- Confirm exact CV: paper uses 25% random train / remainder test, averaged over 5 subjects — is it per-subject or pooled? (Read repo code to confirm.)
- Decide whether the *project* baseline reports MAP-parity or BSC numbers as the headline (recommend: report both; lead with BSC since the RTL is binary).

M1 status: DONE

- [x] Anchor paper selected (Rahimi ICRC 2016)
- [x] Link recorded (ETH PDF + DOI + code repo)
- [x] D, features, window, CV, accuracy captured
- [x] Gaps vs. our binary/1024 RTL flagged
- [ ] (M2) Obtain dataset — next task